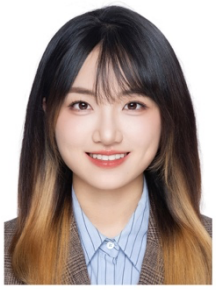
# Yutong Chen

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## RESEARCH INTERESTS

LLM Agents, Agent Harness & Evaluation, Agentic RL, Efficient Model Adaptation, Deepfake Detection

## EDUCATION & ACADEMIC TRAINING

### Education

**University of Cambridge** **Cambridge, UK**
*MPhil in Scientific Computing* Expected start: 10/2026

- **Core Courses:** Mathematical Methods for Scientific Computing, Scientific Programming in C++, Programming with GPUs, Parallelization for HPC.

**Jinan University** **Guangzhou, China**
*BEcon in Economics & Statistics* 08/2022 - 06/2026
*BSc Applied Mathematics with Statistics*

- **GPA: 93/100 | Rank: 5/65**
- **Core Courses:** Linear Algebra (96), Probability & Statistics (94), Applied Statistics (96), Stochastic Process (97), Integer Programming & Combinatorial Optimization (94), Mathematical Finance (93).

### Additional Academic Training

**The Hong Kong University of Science and Technology (Guangzhou)** **Guangzhou, China**
*Visiting Research Student* 10/2025 - 06/2026

- Conducted research on multimodal agents and agentic visual analytics.

**HKU Business School, University of Hong Kong** **Hong Kong, China**
*Global Leader Programme - Business Analysis Track* 07/2023

## HONORS & AWARDS

- China National Scholarship (**Top 0.2%** Nationwide) 10/2025
- China National Entrepreneurship Training Program for College Students (**Top 2%**) 05/2025
- Jinan University Outstanding Student Prize (**Top 1%**) 05/2025
- Xiaomi Scholarship (**1st in each Department**) 12/2024
- Jinan University First Class Scholarship (**Top 3%**) 12/2024
- First-Prize in SPSSPRO National University Students' Data Analysis Competition (**Top 5%**) 08/2024
- Provincial Third-Prize in China Undergraduate Mathematical Contest in Modeling 01/2024
- Bronze Prize in "Internet+" Innovation and Entrepreneurship Competition 07/2023

## PUBLICATION

**Chen, Y.***, Tang, Z.*, Mai, Z., Shuai, Z., Luo, D., Xu, J., Yang, W. "WidgetVA: A Widget-Centric Framework and Benchmark for Agentic Visual Analytics". Under review at *IEEE TVCG*

Shuai, Z.*, Chen, Yiyun*, Ma, M., **Chen, Yutong**, Qiu, H., Xu, J., Chen, Z., Yang, W.† "Learning from Near-Misses: Error-Aware Contrastive Few-Shot Learning for NL2Formula." **Accepted to *ACL 2026 (Main Conference, CCF-A)***

Ye, J.*, **Chen, Y.***, Li, J., Deng, P. "Stock Index Prediction Based On VMD-ARIMA-CNN-LSTM". In Proceedings of *2024 International Conference on Digital Society and Artificial Intelligence* ***(DSAI '24)*****. ACM, pp. 196–205. DOI: 10.1145/3677892.3677927.** * *Equal contribution.*

## RESEARCH EXPERIENCE

**WidgetVA: Agentic Visual Analytics | Co-first Author & Project Leader** **09/2025 - 08/2026**

*Supervisor: Weikai Yang, Tsinghua University & HKUST(GZ)*

- Designed a domain-specific **agent harness** for visual analytics, defining model-facing **state-action-perception** abstraction and implemented WidgetVA Kit, a JavaScript toolkit that exposes heterogeneous VA systems through a unified agent-facing interface.
- Implemented the abstraction with **six widget families**, cross-widget workspace state, provider adapters, and **76 reusable analytical workflows** for closed-loop, **long-horizon interaction.**
- Built **WidgetVABench** with **1,124 human-annotated tasks** and separate Answer, Reference Trace Similarity, and State metrics to diagnose reasoning, tool-grounding, and state-construction failures; controlled evaluation showed multi-widget gains of **5.8/15.1/23.1 points** over direct tools.

**THUNLP AI For Science Research at TsinghuaNLP | Key Member** **04/2025 - Present**

*Supervisor: Zhiyuan Liu & Zheni Zeng, Tsinghua University*

- Reproduced and **DiffSBDD** and Pocket2Mol for protein-pocket-conditioned ligand generation; built reproducible PDB/SDF preprocessing, inference, and evaluation pipelines.
- Evaluated molecular validity, drug-likeness, and ADMET-related properties with **RDKit, QED/logP, and ADMET screening**; performed failure analysis on toxicity and generation quality to guide model selection and experimental iteration.

**Smart Interactive System Based on DeepSeek and QLoRA | Project Leader** **12/2024 - Present**

*Supervisor: Yizun Lin & Xiaoning Zhang, Jinan University*

- Built a **parameter-efficient adaptation** pipeline to train **DeepSeek-R1-Distill-Qwen-14B** on a self-constructed **7,000+ sample** domain dataset using QLoRA; and evaluated generation quality with BLEU/ROUGE.
- Developed a **retrieval-augmented domain agent** with intent routing and hybrid semantic/lexical retrieval, integrating model adaptation and retrieval design for administrative and academic workflows.

## INTERNSHIPS

**Shenzhen Make Block Co., Ltd. | AI Engineer Intern** **07/2025 - Present**

- Built a **LangGraph-based** tool-use agent for xTool laser engravers and used **LangSmith traces** as an observability layer to diagnose tool-selection, state-transition, and multi-step execution failures.
- Owned Planner-side learning in a **Planner-Worker mobile-agent architecture** for 8-12-step e-commerce tasks; LoRA-fine-tuned **Qwen3-8B** on external e-commerce data and internal user logs, establishing an **85% task-completion baseline**.
- Developed a trajectory-based **agentic-RL** pipeline for sparse and delayed rewards: collected **~25K interaction rollouts** to train a Qwen3-8B + lightweight MLP progress estimator for **stepwise credit assignment**, added grounding-based action-validity rewards, and optimized the Planner with **PPO**;
- Improved completion from **85% to 90%**, action-grounding accuracy to **96.2%**, average completion time from **25s to 18s (-28%)**, and user intervention rate to **8%**.

## SKILLS

**Languages & Standardized Tests:** Chinese (native), English (IELTS 7.0); GRE: 324/340, CET-6: 630/710

**Computer skills:** Python: PyTorch, NumPy, Pandas, RDKit; **LLM/Agent/RL:** LoRA/QLoRA, RAG, LangChain, LangGraph, LangSmith, PPO; **Systems:** Linux, Docker, LaTeX.